Recording and fine-tuning guide

VisionICU · research prototype, not for clinical decisions

How to collect data with VisionICU, first in a developer test room and then in an ICU/IMCU, and how that data becomes validated (and, where it makes sense, fine-tuned) models.

> **Read section 1 first.** Every session records numbers once a second, which is enough to **validate** every module. Tick **Capture training data** when starting a session to also record the per-frame inputs the breathing CNNs and the fall model learn from. Face video is never recorded, so EfficientPhys and the pain model cannot be fine-tuned with this yet.

1. What the website records

	Every session	With Capture training data	Needed to fine-tune
Fused RR, HR, early-warning score, pain probability (1 Hz)	Yes	Yes	—
Every estimator's value, quality and raw network output (1 Hz)	Yes	Yes	—
Reference values (*Log reference*, or imported)	Yes	Yes	Yes: RR labels
Marked events (fall, chair exit, talking, CPOT/BPS score, …)	Yes	Yes	Yes: fall labels
Chest / abdomen depth means, every frame	No	Yes	Yes: RR CNN (D435)
Whole-frame mean green, every frame	No	Yes	Yes: RR CNN (LBRD-IC)
Pose keypoints + box, every frame	No	Yes	Yes: fall GRU
Face crops	Never	Never	EfficientPhys, pain model (not supported)
Continuous PPG / respiration waveform	via `scripts/import_reference.py`	same	beat/breath-level labels (future)

Files per session (`data/sessions/<bed>_<time>_<label>/`, git-ignored):

File	Contents
`meta.json`	label, condition, notes, times, fusion setting, `capture`, frame and event counts
`estimates.csv`	1 Hz numbers per estimator
`references.csv`	references typed during the session, plus imported ones (`import:<source>` in the note); the original is kept as `references.manual.csv` after an import
`events.csv`	`t, kind, value, note` for every *Mark an event* press
`frames.csv`	capture only: one row per frame, columns in `src/serving/session_log.py → FRAME_COLUMNS` (~30 rows/s, about 40 MB per hour)
`ppg.csv` / `resp.csv`	an imported reference waveform

Score sessions on the **Compare** page or with `python scripts/compare_sessions.py --by condition --out reports/compare.md`.

2. What each module learns from, and what "fine-tuning" means for it

Module	Has weights?	How to improve it for your setup	Input it needs	Identifiable data?	Label / reference
RR, depth (DSP)	No	Tune `config/signal.yaml` (band, high-pass, segment, gates) against sessions	— (sessions are enough)	No	Metronome / counted breaths / capnography
RR, thermal (DSP)	No	Same; align the thermal camera once per mount	—	No	Same
HR, POS (DSP)	No	Lighting and distance matter more than parameters (README → *Getting a strong signal*)	—	No	Pulse oximeter / ECG HR
RR CNN, D435	Yes (`models/finetuned/rr_cnn1d_d435_finetuned.pth`)	Fine-tune on 10 s chest-depth windows → rate	Chest depth mean, 30 Hz	**No** (numbers)	Breath rate per window
RR CNN, LBRD-IC	Yes (public)	Fine-tune on 10 s green-channel windows → rate (new file in `models/finetuned/`)	Whole-frame mean green, 25 Hz	**No** (numbers)	Breath rate per window
EfficientPhys (HR)	Yes (public)	Fine-tune on face-frame differences → pulse waveform	72×72 face crops, 30 Hz	**Yes (face video)**	Continuous PPG waveform, time-synced
Fall GRU	Yes (public)	Fine-tune on 32-frame keypoint windows → fall / no fall	Pose keypoints, 30 Hz	Low (skeleton only)	Event labels (start/end of each fall, exit, ADL)
Pain (MobileNetV3)	Yes (public, synthetic)	Retrain on real faces with clinical pain scores	Face crops	**Yes (faces)**	CPOT or BPS score by trained nurses
TAA	No	Tune threshold (1.2 rad) and sustain (10 s) against labelled episodes	—	No	Respiratory belts (RIP) or clinician-labelled episodes

Public weights stay untouched; every fine-tuned file goes in `models/finetuned/` and is switched on by pointing its path in `config/signal.yaml` at it.

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3. Before any human recording: approvals (do not skip)

Not legal advice. Confirm every point with your institution.

- ☐ **Institutional Ethics Committee (IEC) approval** for the study protocol (ICMR *National Ethical Guidelines for Biomedical and Health Research*, 2017). One protocol can cover the developer-room volunteer study; the ICU study normally needs its own approval at the hospital.
- ☐ **Clinical Trials Registry – India (CTRI)** registration if the IEC classes the ICU study as clinical research.
- ☐ **Informed consent** forms in English and the local language. ICU patients who cannot consent: consent from the **legally authorised representative (LAR)**, and re-consent from the patient when they regain capacity.
- ☐ **Data protection** under the Digital Personal Data Protection Act, 2023: purpose stated in the consent, pseudonymous subject codes (S01, P01…), the code-to-name key kept on paper or a separate encrypted drive, data on an encrypted disk, never in git or a public cloud, a retention/deletion date.
- ☐ **Hospital permission** (medical superintendent / ICU head) to mount a camera, and a named clinician co-investigator in the ICU.
- ☐ A clear statement to staff that VisionICU is a **research prototype**: no treatment decision may use it.

Numbers-only sessions (section 1) contain no images. Face-video capture for EfficientPhys and pain is identifiable data and needs explicit consent for video.

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4. Equipment checklist

Sensing

- ☐ Intel RealSense D435 + USB 3 cable (≤ 2 m, or an active extension), rigid mount or tripod
- ☐ Waveshare ESP32-S3 thermal camera + 5 V supply (optional; enables the thermal RR path)
- ☐ Laptop or Jetson running `python main.py`; a second screen for the nurse view is useful

References (ground truth)

- ☐ Fingertip **pulse oximeter** with a readable HR (spot values for validation)
- ☐ For EfficientPhys fine-tuning: a **logging PPG sensor** (e.g. a pulse oximeter with PC data export, or a MAX30102 board on an ESP32) recording the waveform at ≥ 30 Hz with timestamps
- ☐ **Metronome app** (phone) for paced breathing
- ☐ For TAA: a pair of **respiratory inductance plethysmography (RIP) belts** (chest + abdomen), if the department has them; otherwise clinician-labelled episodes
- ☐ In the ICU: the **bedside monitor** (ECG HR, impedance or capnography RR). Note its values once a minute, or export them if the monitor allows it
- ☐ One clock source: set the laptop, phone and any logger to network time before each session

Environment

- ☐ Measuring tape and an angle app (camera distance and tilt)
- ☐ A lamp with a diffuser (or a white wall to bounce light off), placed behind the camera
- ☐ For fall recordings (developer room only): crash mats, a spotter, helmet/knee pads offered

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5. Setting up the space

5a. Developer test room (chair)

1. **Chair**: a stable armchair, against a plain wall, nothing hanging behind the subject's head. No window behind the subject (backlight makes the face dark).
2. **Camera**: in front of the chair, **1.5–2.0 m** away (config assumes 1.6 m), lens at **chest-to-face height**, tilted **10–30° down** (config assumes 15°). The face should be at least ~120 px wide in the colour image and the whole torso (shoulders to hips) in view.
3. **Light**: diffuse light falling on the face from behind the camera. Avoid flickering LED strips. Then start the monitor: the camera meters the room at startup and locks exposure, so set the lights first.
4. **Thermal camera** (optional): next to the D435, same direction, about 1 m from the face.
5. **Website**: open the bed → **Set up chair** → click the four seat corners → *Start monitoring*. If thermal is on: **Align thermal** → hot mug at 4–6 spots (5 or more lets it measure its own accuracy).
6. Check on the bed page: face blurred, skeleton on the subject, signal bars above 40 % after ~30 s of sitting still. If not, fix light or distance before recording.

5b. ICU / IMCU (bed or chair)

1. Mount at the **foot of the bed or the side**, 1.5–2.0 m from the chest, high enough to look over bed rails and IV poles, tilted down so face, chest and abdomen are visible.
2. Bed head raised 30–45° is ideal. A blanket over the abdomen is fine; over the chest reduces breathing signal.
3. Nothing hanging between camera and patient (IV lines, curtains); agree with nursing staff where the camera goes so it does not block care.
4. **Set up chair** with *Bed* selected and the four mattress corners. Redo it whenever the bed is moved.

5. Use `--reuse-calibration` only when neither camera nor bed has moved.

6. The training tools

Tool	What it does
Capture training data (session start form)	Writes `frames.csv` as described in section 1. Off by default.
Mark an event (shown while a session runs)	Fall, chair exit, sit-to-stand, lying down, talking, movement, breath-hold start/end, paradoxical breathing start/end, procedure start/end, pain score (CPOT 0–8 or BPS 3–12, the scale is stored at the start of the note), other. Press it **as it happens**, e.g. *Fall* as the person lands.
`scripts/import_reference.py`	Adds a pulse-oximeter or bedside-monitor export (time + hr/rr/spo2) to `references.csv`, or saves a PPG/respiration waveform as `ppg.csv` / `resp.csv`. `--offset-s` corrects the device clock, `--date` reads clock-only times.
`training/dataset.py`	Builds windows exactly like the live modules: 10 s chest depth → 300 samples, detrend, z-score (D435 CNN); 10 s green → 250 samples, z-score (LBRD-IC CNN); 32-frame 62-D pose features replayed through the live fall feature extractor. Skips the first 30 s and 20 s after each reference change, and windows with gaps.
`training/finetune.py`	Subject-wise train/val/test split (written to `data/splits/<module>_splits.json`, checked by `tests/test_leakage.py`), fine-tunes from the current weights with augmentation and early stopping, scores before/after on the held-out people, saves `models/finetuned/<module>_<date>.pth` + a `.json` report. Refuses with fewer than 3 people.
`training/eval.py`	Scores any weights on held-out people; for breathing also scores the signal-processing reading (`rr.depth`, fused `rr`) over the same seconds with MAE **and** coverage.

Subject codes come from the session label: everything before the first underscore (`S07_cabin_paced15_bright` → `S07`). Use the same code for every session of one person.

7. Who to record (subjects)

These are planning targets, not statistical guarantees. Use your IEC protocol's numbers.

Dataset	People	Per person	Spread to cover
Developer validation (numbers only)	10–15 healthy volunteers	30–40 min (protocol in 8a)	Skin tone (Fitzpatrick III–VI matter most in India), sex, age 18–70, beards, glasses, BMI
RR CNN fine-tune (D435 and LBRD-IC)	≥ 20	≥ 20 min of breathing across rates 6–30 /min	Same spread; sitting upright and reclined
EfficientPhys fine-tune	≥ 30 (UBFC-rPPG had 42)	≥ 5 min still + 5 min after light exercise, PPG waveform logged	Skin tone and lighting above all
Fall GRU fine-tune	≥ 20 actors (developer room only)	≥ 10 falls + ≥ 20 non-fall activities each	Fall directions; look-alikes (picking things up, sitting down hard)
Pain model	≥ 50 ICU patients (observational only)	Every routine painful procedure + rest, CPOT/BPS by 2 nurses	Intubated and not; age; skin tone
ICU validation	≥ 20–30 patients	1–2 h each, bedside monitor as reference	Ventilated and spontaneous breathing; bed and chair

Never put the same person in both training and testing.

8. Recording protocols (on the website)

General rules for every session:

- **Record & compare** page → choose the bed → *Condition*, *Label*, reference RR/HR, notes. **Label format:** `S07_cabin_paced15_bright` or `P03_icu_rest`. **No names** in label or notes.
- The first 30 s and 20 s after each reference change are not scored, so hold each block ≥ 2 min.

Naming convention (label = `<subject>_<site>_<block>_<light>`, optional `_r2` for a repeat):

Part	Values	Notes
subject	`S01`–`S99` volunteers, `P01`–`P99` patients	Letters then digits, then _. Training reads the subject from this, so one person keeps one code for life
site	`cabin`, `dev`, `icu`, `imcu`	Where it was recorded
block	`rest`, `paced06`, `paced10`, `paced15`, `paced20`, `paced30`, `hold`, `talk`, `exercise`, `recline`, `taa`, `thermoff`, `fall`	Two-digit paced rates so files sort in order
light	`bright`, `dim`	Light at the time the monitor was started

Examples: `S01_cabin_paced15_bright`, `S03_cabin_rest_dim`, `S05_cabin_fall_bright_r2`. Keep the code-to-name list on paper only; names never go in labels or notes.

- Press **Log reference** every time the truth changes (new metronome rate, new oximeter value).
- One session per block keeps the Compare page readable.

8a. Developer-room protocol per volunteer (~35 min)

#	Block	Condition (website)	Reference to enter	Time
1	Sit still, normal breathing	Rest, normal breathing	Counted breaths/min; pulse-ox HR every 30 s	3 min
2–6	Paced breathing at 6, 10, 15, 20, 30 /min	Paced breathing (metronome)	The metronome rate; pulse-ox HR every 60 s	2 min each
7	Breath-hold 15–20 s, three times	Breath-hold	Note each hold start/end	2 min
8	Talking / small movements	Talking or moving	Counted breaths, pulse-ox HR	2 min
9	Dim light (room lights off, lamp only)	Dim light	As 1	3 min
10	After 2 min of stair stepping	After exercise	Pulse-ox HR every 15 s (it falls fast)	3 min
11	Reclined 30–45°	Other (note "reclined")	As 1	3 min
12	Voluntary paradoxical breathing (belly in while chest rises), only if comfortable	Other (note "TAA")	RIP belts if available, else note start/end	2 min
13	Thermal camera on/off repeat of block 1	Rest, normal breathing	As 1	3 min

Fall block (separate session, developer room, mats + spotter, healthy actors only): forward, sideways and backward slides from the chair to the mat, chair exit, sit-to-stand, bending to pick something up, sitting down hard. Tick **Capture training data** and press **Fall**, **Chair exit**, **Sit-to-stand** etc. under *Mark an event* the moment each happens. For breathing sessions you intend to train on, tick **Capture training data** too.

8b. ICU / IMCU protocol per patient (with consent)

#	Block	Reference	Time
1	Rest, patient settled, no procedures	Bedside monitor RR and HR noted every minute	30–60 min
2	Routine care (turning, suctioning, physiotherapy)	Note start/end; CPOT/BPS before, during, after	As it happens
3	Sitting out in a chair (if the patient does)	As 1; recalibrate as *Chair*	20–30 min
4	Night (lights dimmed)	As 1	30 min

Do **not** stage anything with patients: no paced breathing, no breath-holds, no falls.

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9. From recordings to models

A. Validate (possible today):

1. Record sessions (section 8) with the current models.
2. **Compare** page or `scripts/compare_sessions.py --by condition`: read **MAE together with coverage**, bias, % within ±2 br/min / ±5 bpm, Pearson r.
3. Tune DSP settings in `config/signal.yaml`, re-record on **new** volunteers, compare again.

B. Fine-tune (breathing CNNs and the fall model):

1. Record sessions with **Capture training data** on, references logged (or imported) and events marked. Copy the session folders of every site into one `data/sessions` folder.
2. Import device references if you have them:

   ```
   python scripts/import_reference.py data/sessions/<id> pulseox.csv --source pulse_ox --offset-s 0
   ```

3. Fine-tune (one module at a time):

   ```
   python training/finetune.py --module rr_d435 --sessions data/sessions
   python training/finetune.py --module rr_lbrd --sessions data/sessions
   python training/finetune.py --module fall    --sessions data/sessions
   ```

Read the printed `test before` / `test after` lines: they score people the model never saw.

4. Compare with what the monitor already does:

   ```
   python training/eval.py --module rr_d435 --weights models/finetuned/rr_d435_<date>.pth
   ```

5. If it is better, point `config/signal.yaml` at the new file (`learned.rr_d435.weights`, `learned.rr_lbrd.weights` or `fall.gru_weights_primary`), restart, and record validation sessions on **new** people.
6. Fuse a learned breathing model (`learned.fuse` list) only if it beats the signal-processing path on MAE **and** coverage across conditions. Fused learned paths stay secondary.

EfficientPhys and the pain model need face video, which the recorder deliberately never stores. Supporting them needs a separate, consented and encrypted capture path.

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10. Printable session checklist

Before

- ☐ Consent signed (video consent ticked if capturing faces); subject code assigned
- ☐ Clocks synced; pulse oximeter charged; metronome ready
- ☐ Camera distance 1.5–2.0 m, tilt 10–30°, face ≥ 120 px wide, torso in view
- ☐ Lights set **before** starting `python main.py` (exposure locks at startup)
- ☐ Chair/bed calibrated; thermal aligned (if used); signal bars > 40 % at rest
- ☐ Alarm sound on; staff told it is a research prototype

During

- ☐ One session per block; label `Sxx_site_block_light`
- ☐ **Capture training data** ticked if the session is for fine-tuning
- ☐ **Mark an event** pressed the moment each fall / exit / talking / procedure happens
- ☐ *Log reference* at every change; notes for events (no names)
- ☐ Watch for "Patient moving" / "Face too dark" and fix the cause if it persists

After

- ☐ Stop the session; check row count on the Compare page
- ☐ Copy `data/sessions/<id>/` to the encrypted study drive; record the subject code on the paper log
- ☐ Note anything unusual (lighting change, camera bumped) in the study log